Machine Learning Fundamentals in Python
-
Unsupervised Learning

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Week - 3

Overview

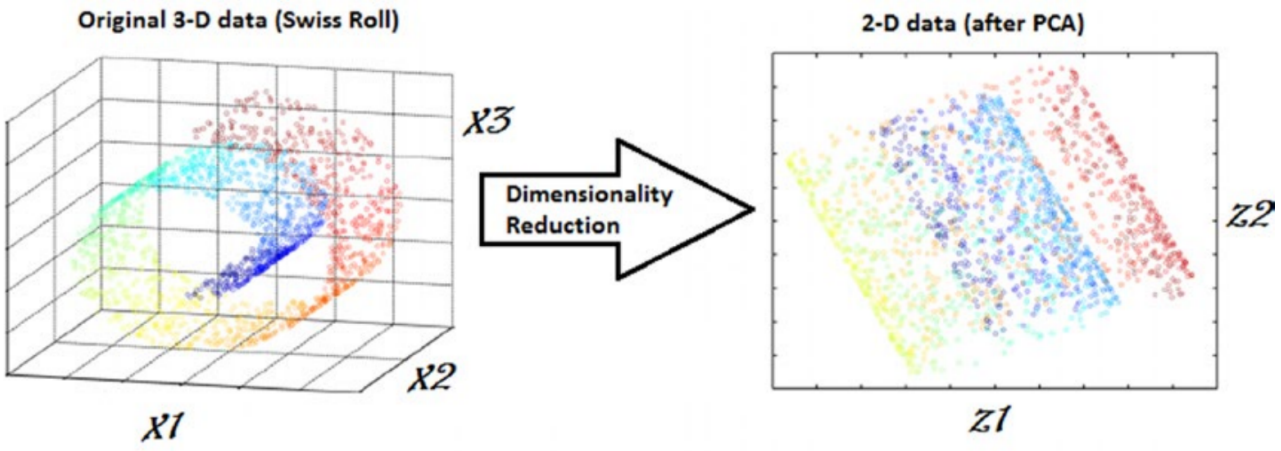
1. Unsupervised Learnign Methods
2. Dimensionality Reduction
3. Clustering

Unsupervised Learning Methods

- Unsupervised learning models or algorithms try to learn inherent latent structures, patterns and relationships from given data without any help or supervision like providing annotations in the form of labeled outputs or outcomes
- Pattern recognition / detection
- Outcome prediction based on previously available supervised training data VS extraction of meaningful insights or information from data
- Can be categorized to the following broad areas of ML tasks:
  - Dimensionality reduction
  - Clustering
  - Anomaly detection
  - Association rule-mining - market basket analysis

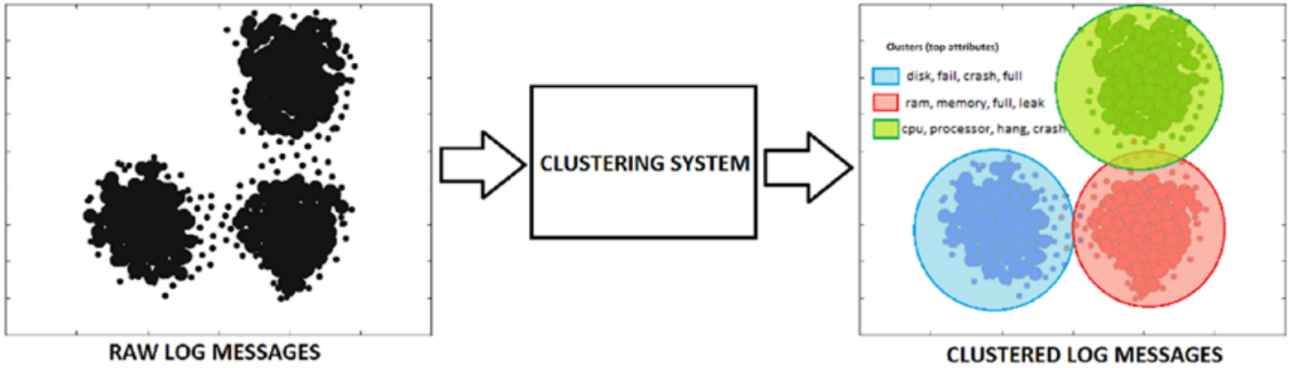
Dimensionality reduction

- Large feature space - multiple challenges - analyzing, visualizing, memory, space constraints - course of dimensionality
- Reduce the number of features of each data sample
- Selecting a set of principal or representative features
  - Feature Selection methods - statistical methods - week 1
  - Feature Extraction methods - extract new features from the original list of features in the data.



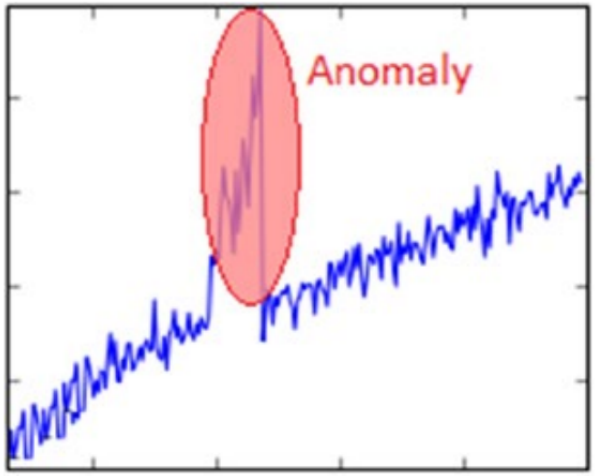
Clustering

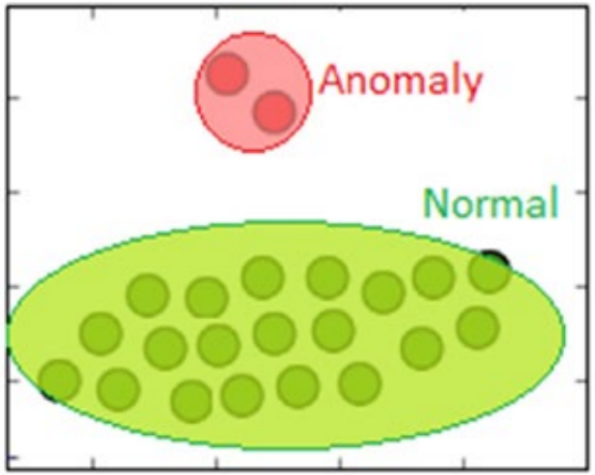
- Find patterns of similarity and relationships among data samples and group data so that each group or cluster of data samples has some similarity, based on the inherent attributes or features
- Real-world problem - multiple servers in a data center and trying to analyze logs for typical issues or errors



Anomaly detection

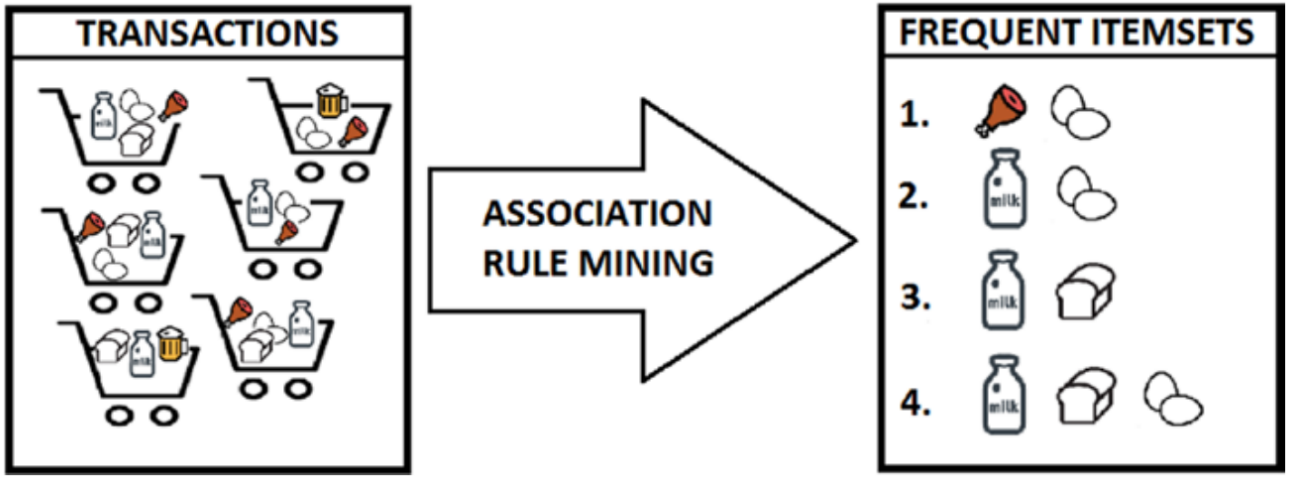
- Outlier detection
- Anomalies occur as infrequently / rare events or as events with specific pattern
- Popular in real-world scenarios like detection of security attacks or breaches, credit card fraud, manufacturing anomalies, network issues, and many more
- Basic idea - train algorithm on not-anomalous data, then any new sample can be identified as normal or anomalous





Association rule-mining - market basket analysis

- Used to examine and analyze large transactional datasets to find patterns and rules of interest
- Association rules help in detecting and predicting transactional patterns based on the knowledge it gains from training transactions
- Helps answering questions like:
  - what items do people tend to buy together = frequent item sets
  - products association - do people who buy beer also tend to buy chicken wings at a pub, etc.



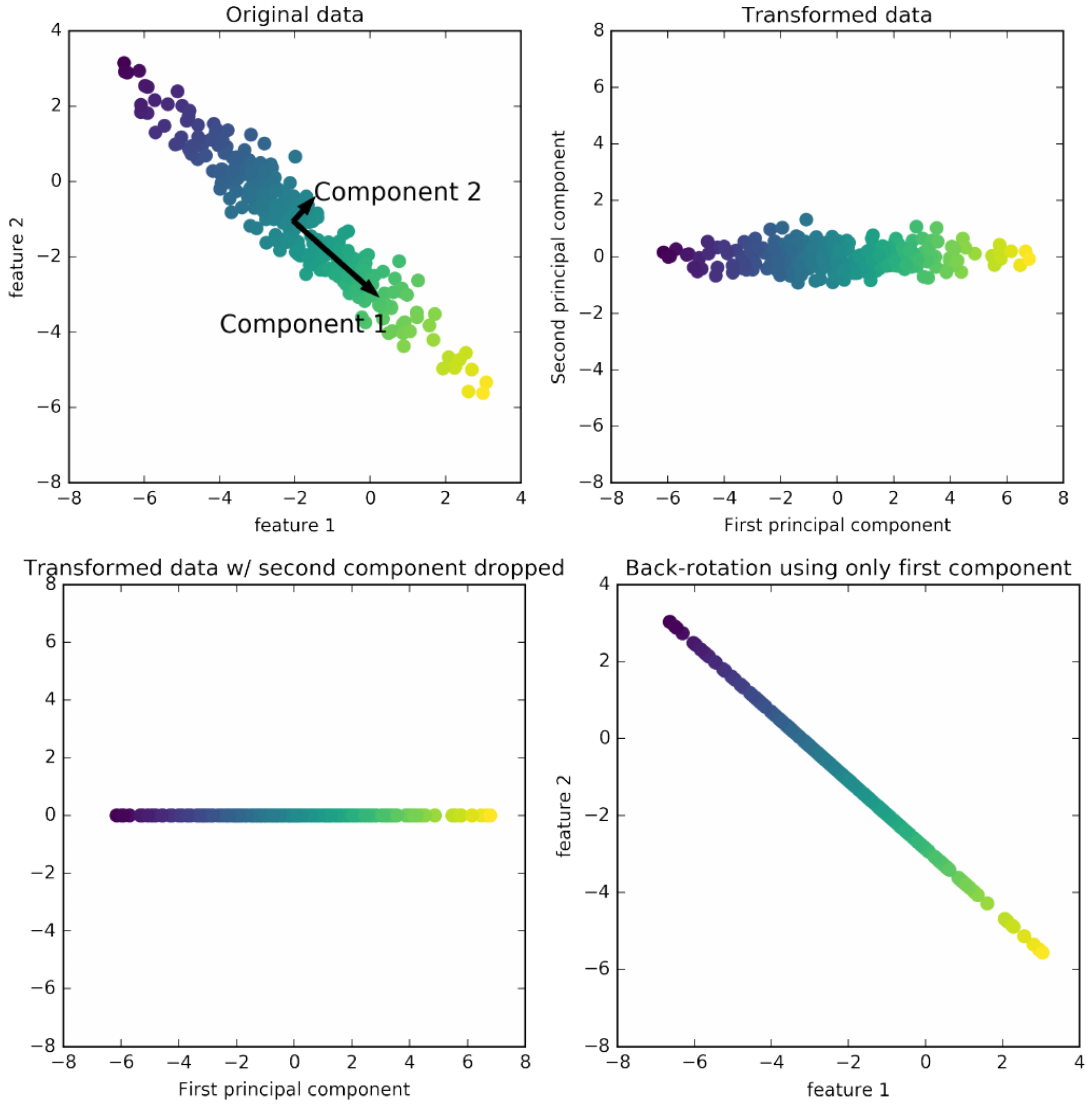
Dimensionality Reduction - PCA

- Principal component analysis (PCA) is a method that rotates the dataset in a way such that the rotated features are statistically uncorrelated.
- The construction of relevant features is achieved by linearly transforming correlated variables into a smaller number of uncorrelated variables
- This is done by projecting (dot product) the original data into the reduced PCA space using the eigenvectors of the covariance/correlation matrix aka the principal components (PCs).
- When to use PCA?
  - PCA technique is particularly useful in processing data where multi-colinearity exists between the features/variables
  - PCA can be used when the dimensions of the input features are high (e.g. a lot of variables)
  - PCA can be also used for denoising and data compression

PCA

- PCA finds a new coordinate system in which every point has a new (x,y) value
- Dimensions do not have physical meaning as they are combinations of original features
- Refered to as "principal components" that are chosen to give lots of variation to one axis
- PCA data = linear combination of original data capturing most of the variance
- PCA is ortogonal transformation - uncorrelated data
- We select subset of the new features, according to how important they are for explaining the data. How?
  - We select features containing most of the variance/information
  - For descriptive purposes, you may only need 80% of the variance explained.
  - Deeper analyses on the data - at least 90% of the variance explained by the principal components.

PCA


Original data
Component 2
Component 1
feature 1
feature 2
Transformed data
First principal component
Second principal component
Transformed data w/ second component dropped
First principal component
Back-rotation using only first component
feature 1
feature 2

PCA vs TSNE

- PCA usually a good first step to transform data so that you can visualize it using a scatter plot
- limitations - linearity between features, preserving global structure by nature, highly sensitive to outliers
- t-distributed stochastic neighbourhood embedding (t-SNE) = non-linear dimensionality reduction and data visualization technique
- Preserves local structures, handles outliers, has hyperparameters = one of the best dimensionality reduction technique

Clustering Methods

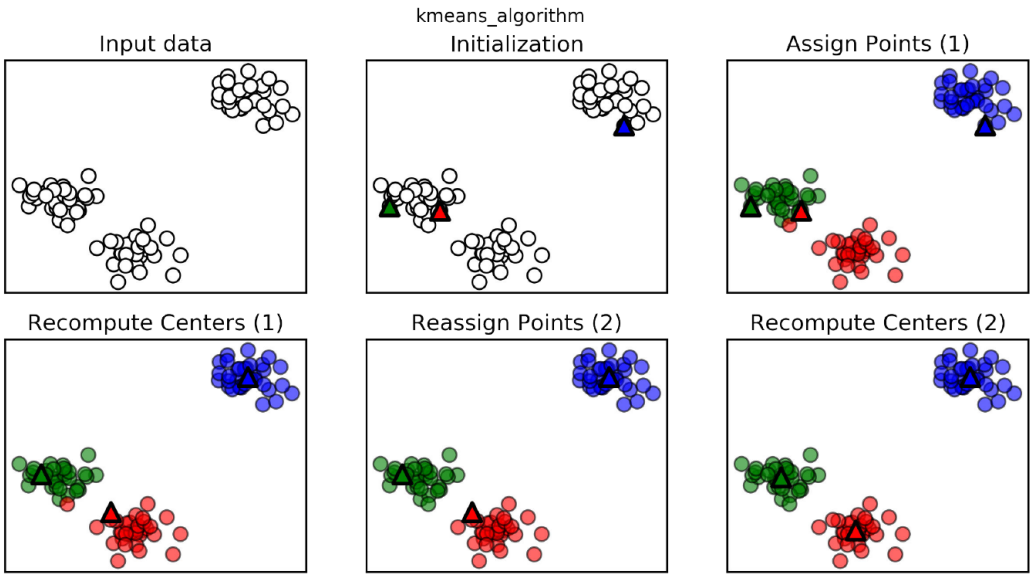
- Types of clustering methods based on following major approaches
  - Centroid based methods such as K-means and K-medoids
  - Hierarchical clustering methods such as agglomerative and divisive (Ward's, affinity propagation)
  - Density based methods such as dbscan and optics
  - Distribution based clustering methods such as Gaussian mixture models

k-Means Clustering

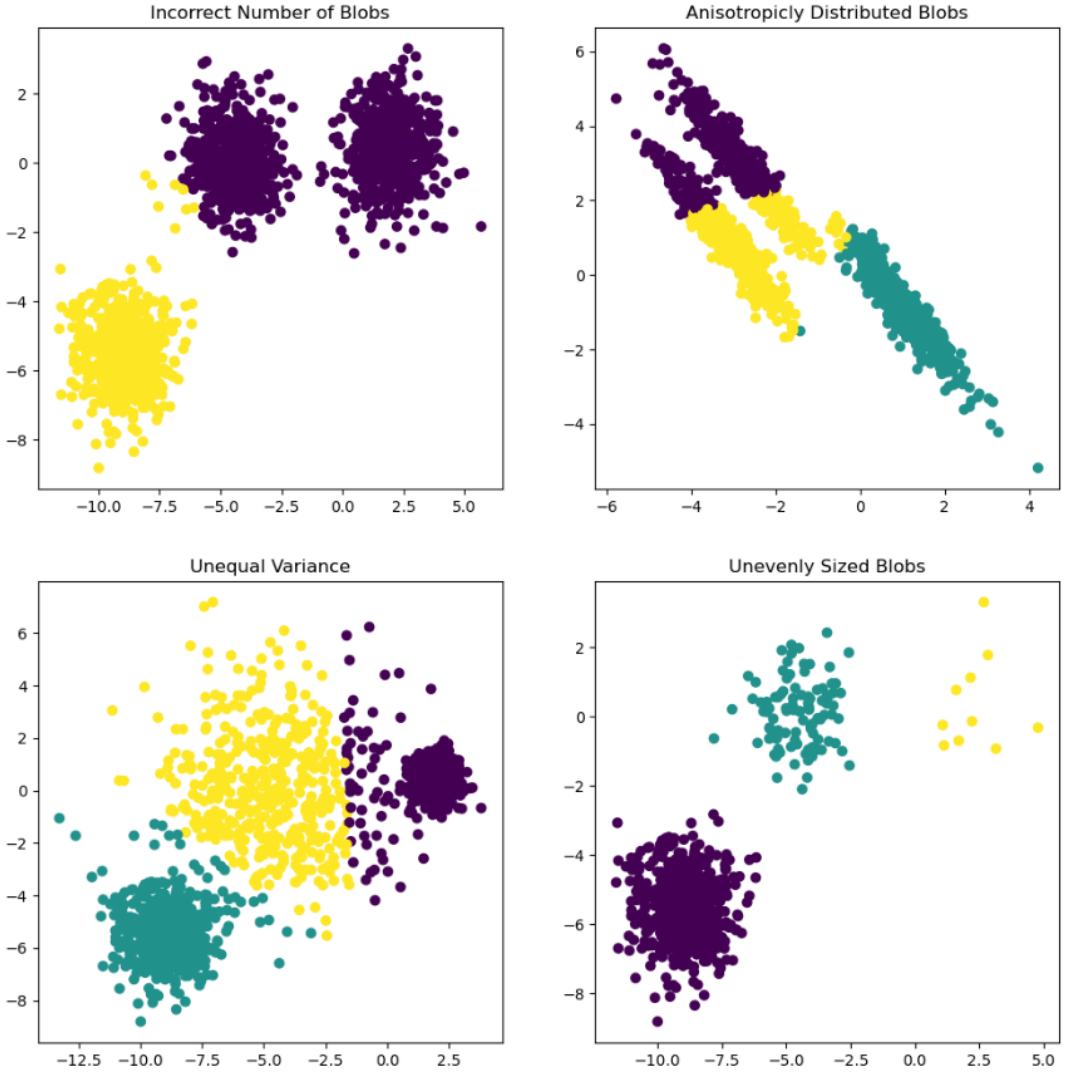
- Simple yet most common method for clustering
- Finds cluster centers representative for regions of data
- The K-means algorithm aims to choose centroids that minimise the inertia, or within-cluster sum-of-squares criterion
- Drawbacks
  - Assumption that clusters are convex and isotropic, which is not always the case. It responds poorly to elongated clusters, or irregular and complex shapes.
  - Inertia is not a normalized metric, we know that 0 is optimal. For higher dimensions - too much time, thus PCA as preprocessing

k-Means Clustering

- Algorithm oscillates between two steps
  - Assign each data point to the closest cluster center
  - Set each cluster center as the mean of the data points that are assigned to it
- Finish when the assignment of instances to clusters no longer changes
- Random initiation, several initiations - select the best in terms of inertia



k-Means Clustering



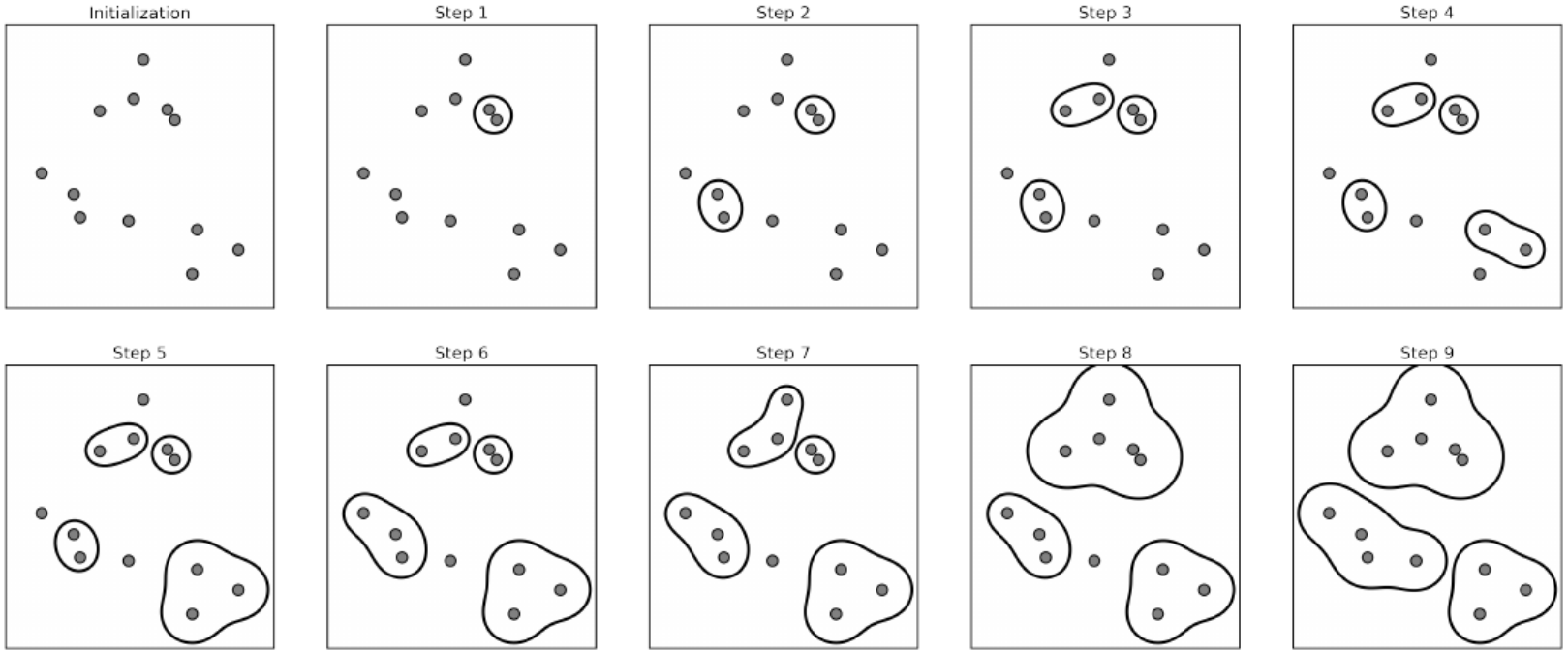
k-Means Clustering Number of Clusters

- Elbow method
  - Calculates the Within-Cluster-Sum of Squared Errors (WSS) for different number of clusters (k) and selecting the k for which change in WSS first starts to diminish
- Silhouette coefficient
  - $S(i) = \frac{b(i)-a(i)}{max\{a(i),b(i)\}}$
  - $b(i)$ is the smallest average distance of point i to all points in any other cluster
  - $a(i)$ is the average distance of i from all other points in its cluster
  - The coefficient tells us if individual points are correctly assigned to their clusters.
  - Rules of thumb while using Silhouette Coefficient:
    - $S(i)$ close to 0 means that the point is between two clusters
    - If it is closer to -1, then we would be better off assigning it to the other clusters
    - If $S(i)$ is close to 1, then the point belongs to the 'correct' cluster
  - Final S is an average of all S(i)

Agglomerative Clustering

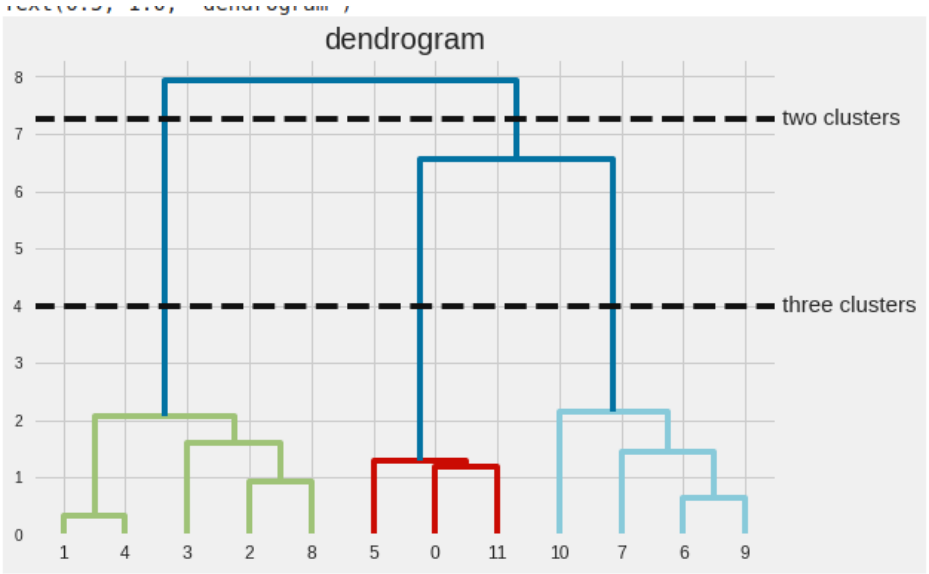
- Same principle - algorithm starts by declaring each point its own cluster, and then merges the two most similar clusters until some stopping criterion is satisfied
- The stopping criterion in scikit-learn is the number of clusters
- How to specify most similar clusters - linkage criteria
  - ward - default - picks the two clusters to merge such that the variance within all clusters increases the least. This often leads to clusters that are relatively equally sized
  - average - merges the two clusters that have the smallest average distance between all their points
  - complete - aka maximum - merges the two clusters that have the smallest maximum distance between their points

Agglomerative Clustering



Hierarchical Clustering and Dendrograms

- Agglomerative clustering = hierarchical clustering
- Clusters created iteratively - from single point cluster to final cluster
- Each intermediate step provides a clustering of the data (with a different number of clusters)
- How to visualize all possible clusterings jointly? Dendogram



DBSCAN

- Density-based spatial clustering of applications with noise
- Does not require the user to set the number of clusters a priori
- Can capture clusters of complex shapes
- Can identify point that are not part of any cluster
- Idea behind - clusters form dense regions of data, separated by regions that are relatively empty
- Two parameters:
  - min_samples
  - eps
- If there are at least min_samples many data points within a distance of eps to a given data point, it's called a core sample. Core samples that are closer than the distance eps are put into the same cluster
- If there are less than min_samples points within distance eps, the given point is labeled as noise
- Init by picking a point to start with